

**FEATURE DRIFT DETECTION TECHNIQUES VIA
ADVERSARIAL APPROACH**

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Thesis
entitled
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ADVERSARIAL APPROACH**

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ABSTRACT

IMPLICATION OF THE THESIS

**KEY WORDS : FEATURE DRIFT DETECTION / ADVERSARIAL VALIDATION /
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CHAPTER I

INTRODUCTION

1.1 Motivation

In real-world AI deployments, models often face data drift or concept drift—gradual or abrupt changes in the data distribution or in the relationship between features and target labels over time [1, 2]. If unchecked, such drift can severely degrade a model’s performance on new data [3, 4]. Failing to detect drift leads to stale models and accumulating technical debt in production systems [5, 6]. It is therefore crucial to detect drift early so that practitioners can retrain or update the model before users are affected [2, 7]. Indeed, waiting until the model’s accuracy drops to react is considered suboptimal, as it means the system has already suffered from poor predictions on new data [1, 3]. Early drift detection is key to maintaining high-quality machine learning services in changing environments [6, 4].

Current drift detection methods are varied. Traditional methods use statistical tests to discover distribution changes by comparing incoming and reference data. Recent comparative studies have reviewed adaptive windowing and incremental distribution-tracking drift detectors, which are still commonly utilized [1]. Modern detectors like KSWIN use the Kolmogorov–Smirnov test to detect distribution changes in streaming data [8]. These statistical detectors actively monitor input distributions (features or model prediction probabilities) and signal drift when data changes.

Some methods track model performance metrics over time. Recent error-based drift detector tests show that cumulative error monitoring and change-point detection still matter for model degradation [3]. The assumption that proper labels will eventually be provided may not hold in many real-time industrial systems.

Recent label-independent proxy approaches identify drift utilizing model outputs or internal signals instead of ground-truth labels. Early drift indications include prediction uncertainty, entropy, and confidence collapse [7]. Comparing model

internal representations between historical and current data detects drift in embedding-space monitoring [4]. Such methods are appealing when labels are limited, delayed, or expensive.

Hybrid drift detection approaches combine multiple sources of information. For example, Context-Aware Drift Detection integrates changes in model predictions with contextual variables—such as user segments and operational metadata—to compute a weighted drift score that reflects distributional changes under different operating conditions [9].

Despite these advancements, existing approaches exhibit limitations when applied in practical environments. Detectors based on statistical tests are frequently susceptible to noise and may generate false alarms. Error-monitoring methodologies depend on the prompt accessibility of ground-truth labels, which presents a considerable challenge in numerous real-world applications. Even the most recent proxy techniques involve significant assumptions or prerequisites. For example, context-aware drift methods depend on the prior definition of appropriate context attributes for conditioning, while uncertainty-based drift indicators (such as those tracking prediction entropy) presume that the model’s output probabilities are accurately calibrated representations of genuine uncertainty. Breaching these assumptions may diminish the efficacy of such detectors. To address these gaps, we propose recasting drift detection as a binary classification problem between reference data (e.g., past/training data) and new data (e.g., current inputs). In other words, if a machine learning model cannot learn to classify accurately whether the data points are from the old or new sets, then the new data is sufficiently similar, and no drift occurs. This approach, called Adversarial Validation, was initially proposed by Pan et al. [10] for feature filtering. However, limited work explores adversarial validation in drift detection settings.

This approach, however, offers several advantages for drift detection: it makes no assumptions about which features or contexts are essential, does not require ground-truth labels for the new data, and provides a single, flexible framework for detecting any type of distributional shift.

1.2 Objectives

The primary objective of this thesis is to investigate adversarial validation as a feature (covariate) drift detection technique for machine learning models deployed in dynamic environments. Specifically, this work aims to:

1. Formulate feature drift detection as a discriminative learning problem by reframing the comparison between historical and incoming data distributions as a binary classification task using adversarial validation.
2. Evaluate the effectiveness of adversarial validation for early feature drift detection under streaming and window-based settings, without relying on ground-truth labels, predefined context variables, or calibrated prediction uncertainties.
3. Systematically benchmark adversarial validation against established drift detection methods, including statistical tests and uncertainty-based proxy metrics, across synthetic and real-world datasets exhibiting different types of drift.
4. Analyze the sensitivity and lead time of detection to determine whether adversarial validation can signal distributional change earlier than baseline approaches.
5. Assess the robustness and practical applicability of the proposed approach in realistic MLOps scenarios, considering factors such as high-dimensional feature spaces, evolving data distributions, and delayed or unavailable labels.

1.3 Expected Contributions

In this thesis, our expected contributions are as follows:

1. Adversarial validation is adaptive as a feature (covariate) drift detection framework by casting drift detection as a discriminative task between historical and incoming data distributions, enabling label-free and model-agnostic monitoring.
2. A systematic benchmark evaluation of adversarial validation for drift detection is conducted, assessing its performance across synthetic and real-world datasets under different drift characteristics and streaming settings—an aspect that has not been explored in prior work.

3. The proposed approach is empirically compared with established drift detection methods, including statistical tests and uncertainty-based proxy metrics, highlighting differences in sensitivity, robustness, and detection behavior.

4. The lead time and stability of adversarial validation–based drift signals are analyzed, demonstrating its potential to provide earlier and more reliable warnings of distributional change than methods that rely on delayed labels or calibrated prediction uncertainty.

5. Practical insights are provided for deploying adversarial validation in MLOps pipelines, including guidance on windowing strategies, threshold selection, and interpretability of drift signals in high-dimensional feature spaces.

CHAPTER II

LITERATURE REVIEW

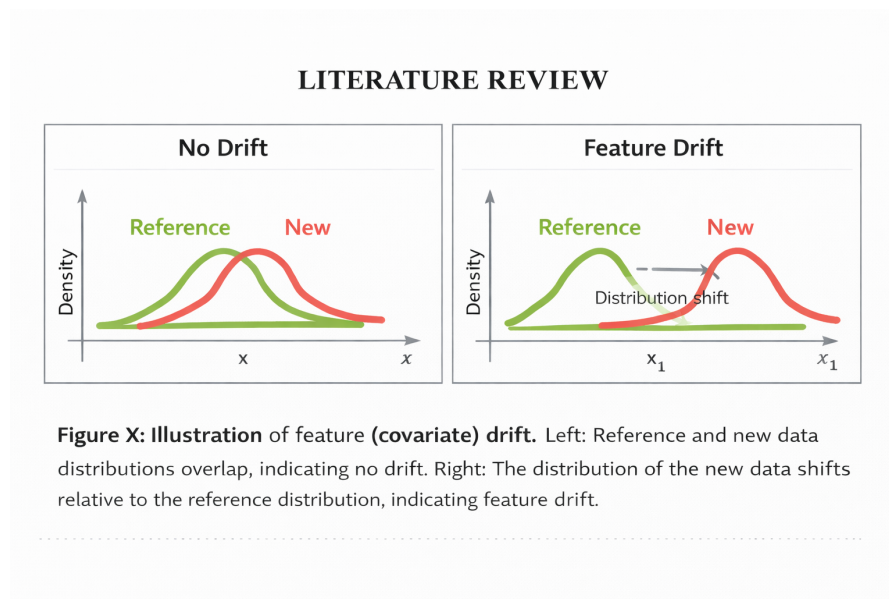


Figure 2.1: Illustration of feature drift.

2.1 Overview

When the statistical characteristics of data evolve over time in a manner that impacts the performance of a machine learning model, this is identified as drift detection. When users, environment, data sources, or the relationships between labels and features change, the model's accuracy diminishes. This modification is indicated early through Drift detection, allowing individuals to adapt, retrain, or update the model accordingly.

2.2 Types of Drift

There are two categories of drift detection:

Concept Drift: The evolving relationship between X and y over time, leading to the deterioration of machine learning model performance. For example, a model trained on past data may experience a decline in performance because current data respond differently, indicating that labels depend on new patterns or rules.

Feature Drift or Covariate Drift refers to the phenomenon where the distribution of the input features changes over time while the relationship between the target variable and the features remains constant [11, 12]. In this case, the ground-truth labels do not change, yet the model's performance may deteriorate because it is exposed to input data that differs from the data seen during training. For example, users may unexpectedly shift their preferences toward different categories of services or products, even though the underlying prediction task remains unchanged [1].

There are various types of feature drift, including:

1. Abrupt drift, where a sudden and rapid change in the feature distribution occurs within a short time period [2];
2. Gradual drift, where a new feature distribution progressively replaces the old one over an extended duration [13];
3. Incremental drift, characterized by small but continuous changes that accumulate into a significant shift over time [1]; and
4. Seasonal or recurring drift, where previously observed feature distributions reappear periodically due to cyclical patterns [14, 2].

Focusing on early detection of feature drift is particularly important in real-world machine learning systems for several reasons. First, feature drift often emerges before observable performance degradation, making it a valuable early warning signal for potential downstream issues [2, 3]. Second, detecting feature drift does not require access to ground-truth labels, which are frequently delayed, expensive, or unavailable in production environments such as recommendation systems, fraud detection, and online advertising [15, 7]. Third, feature drift is especially prevalent in modern data pipelines where user behavior, data sources, and operational conditions evolve continuously [6, 5]. By identifying feature drift at an early stage, practitioners can proactively update data preprocessing, retraining schedules, or monitoring strategies before the model's predictions become unreliable. For these reasons, early detection of feature drift provides

a practical and effective foundation for maintaining robust machine learning systems in dynamic environments.

2.3 Classification of Drift Detection Methods

There are three major classifications that can be used to categorize drift detection:

2.3.1 Analyses of the Statistics

Statistical test-based methods aim to detect drift by directly examining changes in the underlying data distribution over time through hypothesis testing or distance-based measures [2, 1]. Rather than monitoring model error, these approaches focus on the data itself, including input features, predicted probabilities, or summary statistics [16]. A common strategy is to compare a reference data window against a current data window to test whether the two samples originate from the same distribution [17, 18]. Such tests can be applied to numerical and categorical features, learned embeddings, and even projected prediction probabilities [19, 8].

Statistical drift detection methods often employ cumulative statistics or sliding-window mechanisms to capture distributional changes in streaming data [20, 15]. For example, Adaptive Windowing (ADWIN) dynamically adjusts its window size based on detected changes in the monitored statistic, providing theoretical guarantees on change detection [19]. Similarly, KSWIN applies the Kolmogorov–Smirnov test over two sliding sub-windows to identify shifts in data distributions [8]. Other widely used statistical measures include divergence-based metrics such as Kullback–Leibler (KL) divergence, which quantifies the distance between probability distributions and has been applied in various drift detection settings [21, 16].

Overall, these approaches rely on comparisons between historical and recent observations to determine whether statistically significant distributional changes have occurred [2, 3]. While statistical tests are attractive due to their simplicity and label-free nature, they can be sensitive to noise and may struggle in high-dimensional or highly correlated feature spaces [16, 1].

2.3.2 Model Metrics

Methodologies based on model metrics detect drift by systematically monitoring changes in model outputs or predictive performance over time [20, 2, 1]. These approaches are grounded in the assumption that unexpected changes in prediction error characteristics—such as increases in error rate, loss, variance, or cumulative deviation—are indicative of an underlying distributional shift [22, 3]. To evaluate metrics such as classification accuracy, cumulative error, or error variance, these methods typically rely on access to true or delayed labels, which may limit their applicability in real-time production systems [1, 3].

The Drift Detection Method (DDM) detects drift by tracking the classification error rate and its standard deviation over time; when these statistics exceed predefined thresholds, a drift alarm is triggered [20]. The Early Drift Detection Method (EDDM) extends this idea by focusing on the distance between consecutive classification errors, improving sensitivity to gradual changes [22]. The Page–Hinkley (PH) test, as applied by Sebastião and Fernandes [23], employs cumulative sums and two-sided hypothesis testing to identify significant deviations in the monitored signal; exceeding the threshold indicates the presence of drift.

More recent methods such as Hoeffding Drift Detection Method (HDDM) use Hoeffding’s inequality to provide statistically grounded drift decisions. HDDM-A applies this principle to a moving average of prediction errors and classifies the stream state as Stable, Warning, or Drift, while HDDM-W is designed to detect slower, more gradual distributional changes using window-based statistics [24].

2.3.3 Proxy Metrics

Proxy-based drift detection methods are label-independent approaches that infer distributional shifts by monitoring indirect signals such as model outputs, uncertainty estimates, and latent representations, rather than relying on ground-truth prediction errors [16, 1]. These methods are particularly advantageous in real-world systems where labels are delayed, sparse, or unavailable, such as fraud detection, recommendation systems, and click-through rate prediction [5, 7]. Common proxy indicators include prediction confidence drift, output entropy, embedding distribution shift, changes in

feature relevance, and shifts in model output distributions [4, 25].

Context-Aware Drift Detection [9] applies conditional two-sample testing to determine whether the distribution of new data differs from historical data within predefined contexts or subpopulations. This method computes a context-weighted drift score using conditional Maximum Mean Discrepancy (MMD), enabling more localized drift detection [26]. Similarly, embedding-based drift detection techniques compute statistical distances—such as cosine similarity, Wasserstein distance, or energy distance—between reference and current embedding distributions to identify representational shifts [16, 4]. Recent advances leverage representation learning and contrastive objectives to enhance separability between historical and novel concepts; for example, the MCDD algorithm employs contrastive learning to improve drift sensitivity in embedding space [27].

Because proxy metrics do not require labeled data, they enable earlier detection of covariate and concept drift, making them particularly suitable for online, streaming, and black-box machine learning systems [2, 3]. Proxy metrics can be broadly categorized as follows.

2.3.3.1 Prediction-Based Proxy Metrics

Prediction-based proxy metrics detect drift by monitoring changes in model outputs over time, without evaluating prediction correctness [16]. These methods track signals such as prediction confidence, entropy of softmax outputs, class distribution shifts, internal representation changes, and output consistency [28, 25]. Such indicators often reveal distributional change before labeled performance degradation is observed [7].

For example, Prediction Uncertainty Drift monitors increases in predictive entropy or uncertainty, under the assumption that rising uncertainty reflects exposure to unfamiliar data [29]. Another widely used signal is Embedding Distribution Shift, where stability of learned representations is assessed across time; this idea is grounded in domain shift and representation learning literature [30, 31].

2.3.3.2 Data Pipeline / Database Proxy Metrics

Data pipeline-level proxy metrics detect drift by monitoring feature quality, volume, freshness, and integrity upstream of model inference [32, 33]. Typical signals include increases in missing values, shifts in feature value distributions,

invalid categories, out-of-range values, and anomalous temporal patterns such as future timestamps [32]. These metrics act as early indicators of upstream data drift before model-level degradation becomes apparent [5, 16].

2.3.3.3 Operational or System Metrics

Operational or system-level metrics capture drift indirectly by observing changes in deployment-level signals such as latency, error rates, conversion rates, click-through rate (CTR), and other business KPIs [34]. Sudden declines in engagement, conversion, or fraud detection rates may indicate that the underlying data distribution has shifted, even when traditional ML metrics remain stable [33]. Latency-based drift indicators further capture changes caused by upstream pipeline issues, new user segments, or altered input characteristics [35].

2.3.3.4 Hybrid or Composite Metrics

Hybrid or composite approaches integrate multiple proxy signals—such as prediction statistics, embeddings, operational KPIs, and metadata—into a single drift score [2, 3]. For example, Context-Aware Drift Detection [9] combines model prediction behavior, user context, operational signals, and metadata (e.g., device type, location, time) to compute a context-weighted drift score. Such hybrid approaches are particularly effective in large-scale business systems, including fraud detection, financial scoring, and e-commerce personalization [5, 9].

2.4 Limitations of Existing Methods

The literature offers a range of concept drift detection methods, but a detailed review reveals specific gaps that motivate our work. Although the newest techniques have improved drift monitoring, each comes with restrictive assumptions or shortcomings. Context-aware drift detection methods attempt to improve sensitivity by conditioning drift tests on auxiliary variables such as user segments, device types, time-of-day, or operational metadata [9]. While effective in controlled settings, these approaches impose a strong practical requirement: the relevant contextual variables must be identified, engineered, and maintained correctly before deployment. In real production systems—such as recommendation platforms, fraud detection pipelines, or ad-ranking systems—context

is often incomplete, noisy, or evolves over time [33, 6]. For example, a recommender system may segment users by geography or device, yet drift may be driven instead by subtle behavioral changes, platform UI updates, or latent preference shifts that are not captured by predefined context features. When context is misspecified or becomes stale, context-aware detectors may either fail to detect drift or generate spurious alarms triggered by benign contextual fluctuations.

Uncertainty-based drift detection methods, including prediction entropy or PU-index–style indicators [29], face a different but equally restrictive assumption they rely on the model’s output probabilities being well calibrated. However, extensive empirical evidence shows that many modern deep learning and ensemble models deployed in production are poorly calibrated, particularly under distribution shift or class imbalance [28, 25]. In such cases, changes in prediction confidence may reflect overconfidence, underconfidence, or optimization artifacts rather than genuine data drift. For instance, a fraud detection model retrained periodically may exhibit stable entropy levels even as transaction patterns shift, because calibration degrades silently over time. As a result, rising or falling uncertainty does not always correlate reliably with distributional change.

Crucially, neither context-aware nor uncertainty-based approaches explicitly test whether new data is statistically distinguishable from historical data. That is, they do not directly assess the separability between reference and current data distributions. This omission is significant from a practical MLOps perspective: the ability to discriminate between “old” and “new” data provides a general, model-agnostic signal of drift that does not depend on predefined context variables, ground-truth labels, or calibrated confidence estimates. Discriminative tests capture high-dimensional, non-linear shifts that are common in modern ML pipelines but difficult to detect using marginal statistics or proxy metrics alone. Despite its conceptual simplicity and industrial relevance, this discriminative perspective on drift detection has received limited attention in the drift detection literature. Addressing this gap motivates the adversarial validation framework proposed in this thesis, which reframes drift detection as a direct test of separability between past and current data distributions.

Our proposed adversarial validation framework is designed to fill this gap by directly measuring how well one can differentiate new data from old – a capability

that sidesteps the above limitations and provides a more flexible detection criterion. Detecting changes can be complicated. High-dimensional, non-linear feature discrepancies without requiring predefined context variables, error-rate changes, or model-dependent uncertainty estimates.

2.5 Related Work

2.5.1 Adversarial Validation Overview

Adversarial validation was originally proposed and popularized as a practical technique for diagnosing dataset shift between training and evaluation data, by reframing the comparison of two datasets as a binary classification problem. The idea is to train an auxiliary classifier to predict whether each instance originates from a reference distribution (e.g., historical or training data) or from a target distribution (e.g., test or deployment data). If the classifier achieves high discriminative performance, the two datasets are statistically separable, indicating a significant distributional shift; conversely, near-random performance suggests that the datasets are largely indistinguishable. This approach was first introduced in industrial machine learning pipelines to identify train–test mismatch and guide data selection, rather than as a standalone drift detection mechanism.

2.5.2 Industrial Applications

A prominent industrial application of adversarial validation was presented by Pan et al. [10] at Uber, where the method was employed within the MaLTA user targeting automation system. In this setting, adversarial validation was used to filter features and training samples by training a classifier to distinguish historical training data from newly observed targeting data. When the adversarial classifier exhibited strong separability, it signaled a mismatch between past and current data, prompting adjustments to the feature set and retraining strategy before deployment. Their results demonstrated that adversarial validation improved robustness in large-scale production systems compared to naive retraining strategies that react only after observable performance degradation.

Outside of Uber, adversarial validation has been used mainly to manage dataset shift and domain mismatch, not as a dedicated drift detector. Qian and Rao

[36] propose “Managing Dataset Shift by Adversarial Validation for Credit Scoring”, where an adversarial classifier is trained to distinguish historical credit data from more recent application data. Only those training samples whose distribution is closest to the target (deployment) period—identified via low adversarial separability—are kept for cross-validation and model training, improving stability and generalization in changing credit environments. Renza and Moya-Albor [37] apply adversarial validation to image classification datasets, using an adversarial classifier between training and test images to quantify dataset complexity and shift. Their method reports an “adversarial validation score” that reflects how easily a model can discriminate between the two splits, and they use this score to reason about dataset construction and robustness rather than to drive an online drift alarm. In operations management, Senoner et al. [38] employ adversarial validation when addressing distributional shifts in order-picking systems. They train gradient-boosted trees to distinguish samples from different operating conditions, using the adversarial classifier’s performance as an indicator of when the historical data is no longer representative of a new operational regime. However, their use of adversarial validation is embedded inside a broader decision-making and domain-adaptation pipeline, not exposed as a standalone monitoring component.

2.6 Gap in the Literature

Although adversarial validation has been successfully applied to measure dataset shift and improve robustness in specific application-driven settings (e.g., Uber’s MaLTA system, credit scoring, image classification, and operations management), it has not been developed into a general, label-efficient feature (covariate) drift detection method. In particular, adversarial validation has not been systematically studied or evaluated as a standalone detector on standard drift benchmark datasets under streaming or window-based evaluation protocols. Existing uses are typically embedded within domain-specific pipelines rather than framed as general-purpose drift monitoring techniques, leaving its empirical behavior and reliability in benchmark drift scenarios largely unexplored.

CHAPTER III

METHODOLOGY

3.1 Adversarial Validation as Feature Drift Detection

We propose Adversarial validation as a tool for feature drift detection aimed at overcoming the limitations noted above. Its purpose is to proactively detect distribution shifts by leveraging a predictive model as a kind of “drift sensor.” Instead of waiting to observe a drop in the primary model’s accuracy, adversarial validation seeks to flag drift as soon as the characteristics of incoming data begin to differ from past data.

This research follows a streaming drift detection methodology inspired by the Prediction Uncertainty (PU) framework, while introducing adversarial validation as the core mechanism for early feature drift detection. The overall design preserves PU’s window-based comparison, statistical decision-making, and adaptation strategies, ensuring fair comparison and methodological consistency, while replacing uncertainty-based signals with a discriminative, feature-level drift indicator.

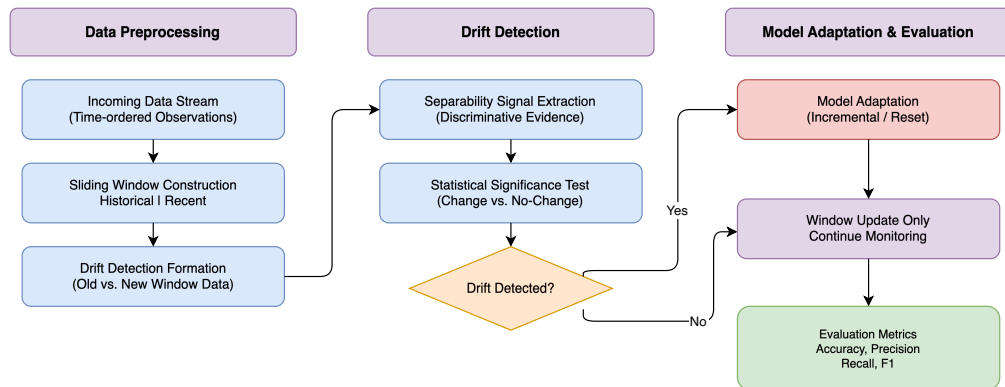


Figure 3.1: Overview of the proposed adversarial validation framework for feature drift detection.

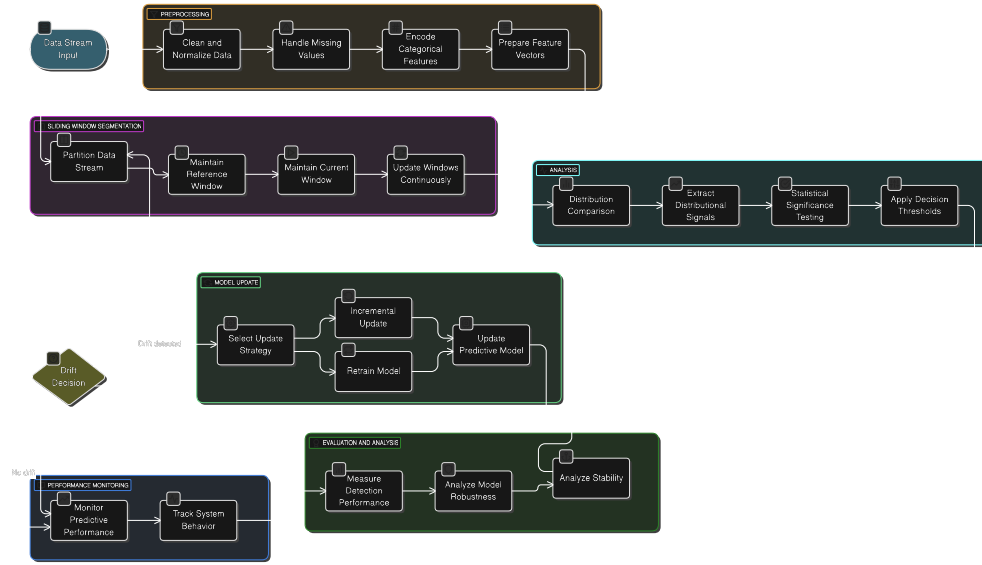


Figure 3.2: Detailed steps of the proposed adversarial validation framework for feature drift detection.

3.2 The core components are:

3.2.1 Datasets

To evaluate the proposed approach, experiments are conducted on a combination of synthetic and real-world benchmark datasets commonly used in drift detection research. Synthetic datasets are employed to simulate controlled feature drift scenarios—such as abrupt, gradual, incremental, and recurring drift—allowing precise analysis of detection timing and sensitivity. Real-world datasets are used to assess the robustness and practical applicability of the proposed method under realistic, noisy, and high-dimensional conditions.

All datasets are processed in a streaming fashion, where instances arrive sequentially over time. Feature vectors are used as inputs for drift detection, while labels are reserved solely for downstream performance evaluation and are not required by the proposed drift detection mechanism itself. This setup reflects real deployment environments where labels may be delayed or unavailable.

3.2.2 Adversarial Validation as Feature Drift Detection

The core contribution of this work is the use of adversarial validation as a feature (covariate) drift detector. Instead of monitoring prediction uncertainty as in the PU approach, drift is detected by measuring the separability between historical and newly observed feature distributions.

At each time step, data are partitioned into two sets:

1. a reference window, representing historical or stable data, and
2. a current window, representing newly arriving data.

An auxiliary binary classifier is trained to distinguish samples from the reference window and the current window using only feature values. The classifier's discriminative performance—measured using metrics such as accuracy or AUC—serves as a proxy signal for feature drift. High separability indicates that the feature distributions differ significantly, while near-random performance suggests distributional stability.

This approach is label-independent, model-agnostic, and capable of capturing high-dimensional, non-linear distributional shifts. By focusing on feature-level separability, it enables early detection of drift before observable performance degradation occurs, aligning with the primary objective of this research.

3.2.3 Experimental Setup

Sliding Window Strategy Consistent with the PU methodology, a sliding window mechanism is used to manage streaming data. Two windows are maintained:

1. a reference window containing historical data, and
2. a target window containing the most recent data.

As new instances arrive, the windows slide forward in time, discarding outdated samples to prevent obsolete distributions from influencing drift detection. This design allows continuous monitoring while controlling memory usage and reducing false alarms caused by stale data. The window sizes and step intervals are kept consistent with the PU framework to ensure methodological comparability.

3.2.4 Model Adaptation Strategies

Following the PU paper, two learning strategies are evaluated to study the interaction between drift detection and model updating:

Training Only at Initialization or After Drift Detection: The model is trained once at initialization and is only retrained when a drift alarm is triggered. This setup highlights the practical value of early drift detection by limiting unnecessary retraining.

By combining these strategies with adversarial validation–based drift signals, the experiments examine how early feature drift detection influences downstream model performance under different adaptation regimes.

3.3 Evaluation

The proposed approach is evaluated using standard drift detection and performance metrics, following the PU evaluation protocol to enable direct comparison.

Predictive Performance Metrics: Classification or regression performance metrics (e.g., accuracy, F1-score, RMSE) are computed using ground-truth labels to evaluate the impact of drift detection on model effectiveness.

Comparative Analysis: The adversarial validation–based method is compared against baseline drift detection approaches, including statistical tests and uncertainty-based proxy methods, under identical experimental conditions.

This evaluation framework enables systematic analysis of early detection capability, robustness, and practical utility, demonstrating the effectiveness of adversarial validation for feature drift detection in streaming machine learning systems.

CHAPTER IV

RESULTS

CHAPTER V

DISCUSSION

CHAPTER VI

CONCLUSION

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